

EXAM 7 – SPRING 2013

15. (1.5 points)

Identify and briefly discuss three tools that can be used in an internal model to quantify the effect of extreme events on an insurer's solvency.

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Question 15 Sample Answer

Solution 1

- (1) CAT models – Used to measure multiple CAT risk, as well as assist in purchase of reinsurance. Help quantify magnitude of extreme event under various scenarios
- (2) Expert Opinion – Draw on knowledge of industry experts to gain insight as to company's exposure to extreme events and what they can do to migrate. This can be subject to problems of inherent use of judgment or be biased
- (3) Economic Scenario Generators – Used to generate different scenarios of inflation rates and other economic factors that can quantify impact extreme event have

Solution 2

- (1) Can explicitly use Cat Models to model the effect (loss distribution) of natural and man-made disaster on the insurance company's losses
- (2) Can use economic scenario generators to simulate different economic scenarios (inflation, risk of credit default, equity /property risks) and how they effect solvency in different scenarios
- (3) Can use Copulas to combine the loss distribution of multiple lines of business (auto, home) and specify how correlated are in different parts of the distribution (tail, for example)

Examiner Comment

Many candidates listed three of the six items from Brooks, et. al. Other answers were accepted as long as consideration was given to "fattening the tail" of the underlying severity distribution. Copulas were also accepted as long as they properly referenced the correlation of lines of business during extreme events.

Often candidates did not provide an explanation of why the tool was relevant to extreme events.